PUSHFORWARD AND PULLBACK ALONG A CLOSED IMMERSION OF ALGEBRAIC SPACES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-morphisms.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted statement is a two-part sheaf-theoretic lemma; the proof handles the fully faithful part and the essential image part in one environment, the second by the same descent device as the first.

Editorial note. The three references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: the three native excerpt spans of sources/stacks-spaces-morphisms/preamble.tex (lines 8-13, 14-18 and 154-217). The chapter file itself carries no preamble text.

Allowed standard prerequisites (editorial).

- Base change formula for the small pushforward along an etale morphism. Properties of Spaces, Lemma spaces-properties-lemma-pushforward-etale-base-change, used at proof lines 2583-2590. Reduces the sheaf statement on X to the corresponding statement on U.
- The scheme case for a closed immersion of schemes. Etale Cohomology, Proposition etale-cohomology-proposition-closed-immersion-pushforward, used at proof lines 2591-2595 and 2602-2605. Is the input that the etale cover pulls back.
- Full faithfulness of a functor between sheaf categories detected on a covering. Sites, Lemma sites-lemma-exactness-properties, used at proof lines 2598-2599. Upgrades the isomorphism on V to full faithfulness on X.
- Descent of the property of being an isomorphism along a surjective etale morphism. used at proof lines 2596-2598 and 2609-2612. Turns the pointwise isomorphisms into global ones.

Source reference. Original proposition 04CA: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 5684–5684. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 04D6: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 5716–5716. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 03LR: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4723–4723. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Date: stacks-project:spaces-morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.2. *Let S be a scheme. Let $i : Z \to X$ be a closed immersion of algebraic spaces over S.*

(1) *The functor*
$$i_{small,*} : Sh(Z_{étale}) \longrightarrow Sh(X_{étale})$$
is fully faithful and its essential image is those sheaves of sets $\mathcal{F}$ on $X_{étale}$ whose restriction to $X \setminus Z$ is isomorphic to $*$, and

(2) *the functor*
$$i_{small,*} : Ab(Z_{étale}) \longrightarrow Ab(X_{étale})$$
is fully faithful and its essential image is those abelian sheaves on $X_{étale}$ whose support is contained in $|Z|$.

In both cases i_{small}^{-1} is a left inverse to the functor $i_{small,*}$.

Proof. Let U be a scheme and let $U \to X$ be surjective étale. Set $V = Z \times_X U$. Then V is a scheme and $i' : V \to U$ is a closed immersion of schemes. By Properties of Spaces, Lemma 03LR for any sheaf $\mathcal{G}$ on Z we have
$$(i_{small}^{-1} i_{small,*}\mathcal{G})|_V = (i')_{small}^{-1} i'_{small,*}(\mathcal{G}|_V)$$
By Étale Cohomology, Proposition 04CA the map $(i')_{small}^{-1} i'_{small,*}(\mathcal{G}|_V) \to \mathcal{G}|_V$ is an isomorphism. Since $V \to Z$ is surjective and étale this implies that $i_{small}^{-1} i_{small,*}\mathcal{G} \to \mathcal{G}$ is an isomorphism. This clearly implies that $i_{small,*}$ is fully faithful, see Sites, Lemma 04D6. To prove the statement on the essential image, consider a sheaf of sets $\mathcal{F}$ on $X_{étale}$ whose restriction to $X \setminus Z$ is isomorphic to $*$. As in the proof of Étale Cohomology, Proposition 04CA we consider the adjunction mapping
$$\mathcal{F} \longrightarrow i_{small,*} i_{small}^{-1} \mathcal{F}.$$
As in the first part we see that the restriction of this map to U is an isomorphism by the corresponding result for the case of schemes. Since U is an étale covering of X we conclude it is an isomorphism. □